USER MANUAL



1. Product Info

Model: Leishen Intelligent N10

Description: LiDAR

Batch Number

2. MANUFACTURER & EU REPRESENTATIVE INFO

Manufacturer: WitMotion Shenzhen Co., Ltd

EU Authorized Representative: Guangming Yunli Intelligent Park, No. 143 Xihuan Avenue, Guangming District, Shenzhen City, Guangdong Province, China

EMAIL: wit@wit-motion.com

Website: http://www.wit-motion.com/

Chinese Website: http://www.wit-motion.cn

Tel: +86-755-33185882

3. Safety & Warnings

CAUTION

Avoid looking directly into the beam.

Read these instructions carefully and retain them for future safety reference.

Do not disassemble or modify the internal sensor components arbitrarily.

This product is intended to be used in dry indoor areas only, avoid damp/wet environment.

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DANGER

Laser radiation, avoid direct eye exposure, improper use may cause irreversible eye injury, keep away from children and pets.

Read all instructions before use to ensure safety and prevent damage.

Warning 1: Laser Safety

This product uses 905nm infrared laser and complies with IEC 60825-1 Class 1 Laser Safety Standard.

Warning 2: Installation & Environment

Do not block the optical window; avoid direct strong sunlight or high reflective mirror surfaces. Keep away from corrosive liquid, heavy dust and impact.

Warning 3: Children Safety

Keep the LiDAR, wiring harness and accessories out of reach of children under 3 years old.

Warning 4: Prohibition

Unauthorized disassembly, modification or optical coating will void all warranty and break laser safety compliance.

4. Product Overview

The N10 series LiDAR adopts TOF (Time of Flight) technology and can perform 360-degree 2D scanning detection of the surrounding environment. The LiDAR uses wireless power supply and optical communication internally, with a measurement repetition rate of 4.5KHz. The detection accuracy reaches +/-3cm with a maximum range of 12 meters.

Main applications: indoor service robots, AGV, cleaning and disinfection robots, drones, and other precise positioning and obstacle avoidance scenarios.

Product Features:

Maximum measurement range: 0m to 12m (70% reflectivity target).

Adopts direct Time of Flight (dToF) ranging principle.

360-degree scanning range, angular resolution 0.48 to 0.96 degrees.

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Coaxial design, compact size, suitable for built-in installation.

5. Technical Specifications

5.1 Performance Parameters

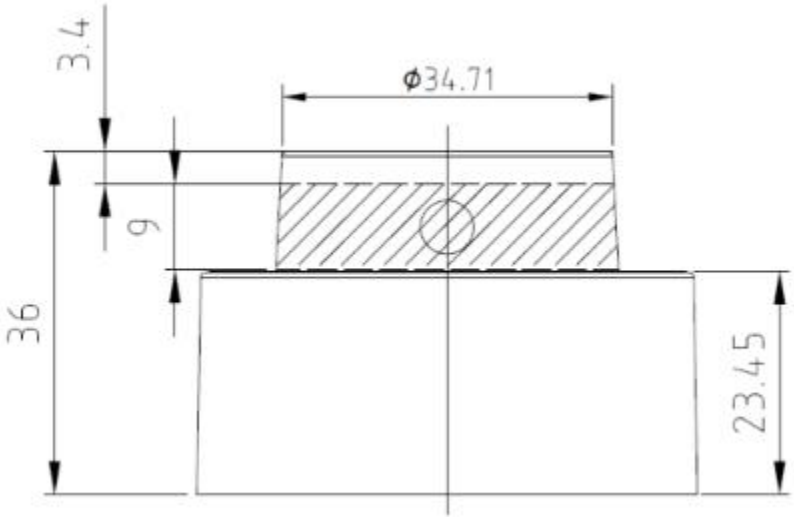
Item	**Specification**
Model	N10
Type	Short Range
Ranging Principle	TOF, supports indoor and outdoor environments
Scanning Angle	360 degrees
Angular Resolution	0.48 to 0.96 degrees
Emission Repetition Rate	4.5KHz
Scan Frequency	6~12Hz
Dimensions	Diameter 52 x 36.1mm
Weight	Approximately 60g
Protection Level	IPX-4
Output Data Resolution	15mm
Measurement Accuracy	+/-3cm (0~6m); +/-4.5cm (>=6m) (70% reflectivity target)
Light Source	905nm Laser
Range	0m to 12m (70% reflectivity target)
Ambient Light Resistance	4K Lux
Power Supply	5V DC (4.75V~5.25V)
Operating Temperature	Work: -10 to 40 C; Storage: -30 to 70 C
Motor Drive	Built-in brushless motor
Communication Interface	Standard UART (Baud rate: 230400bps)

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5.2 Optical Window

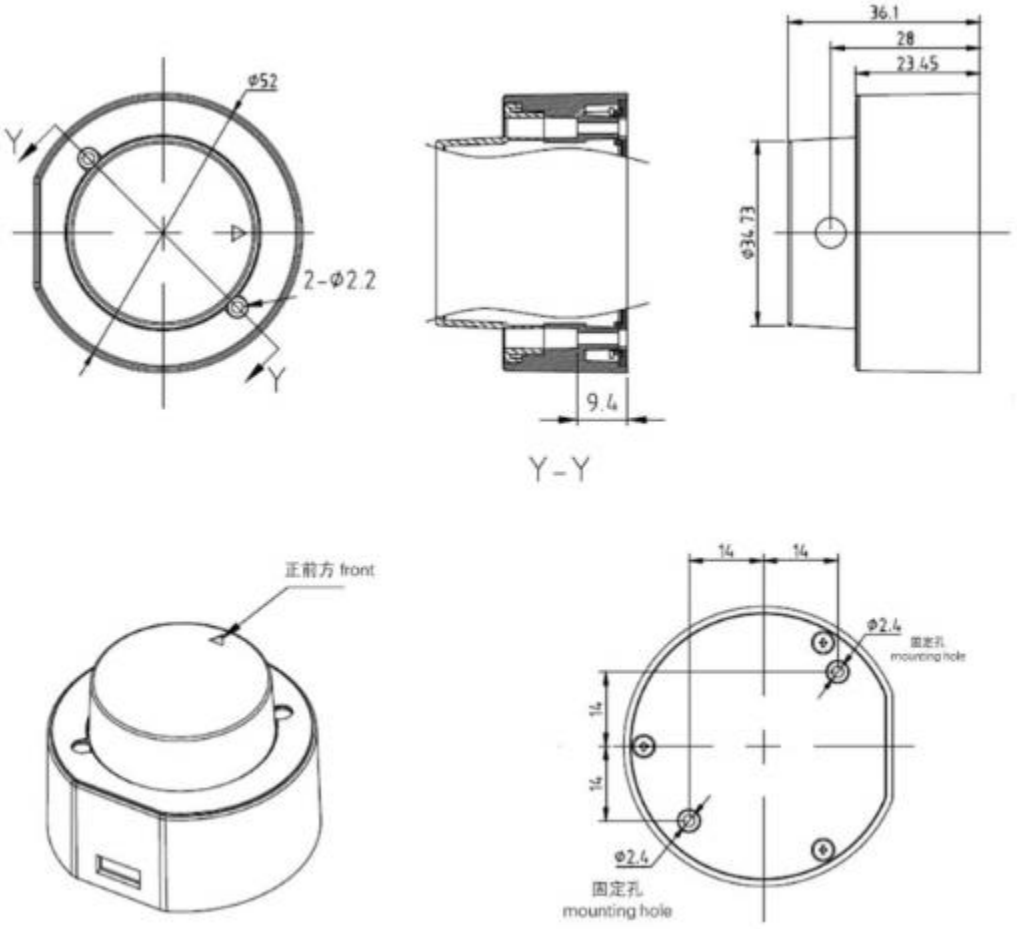
To avoid dust affecting the LiDAR ranging performance, please keep the product optical window clean.

To ensure normal operation, avoid blocking the optical window when designing the external system.



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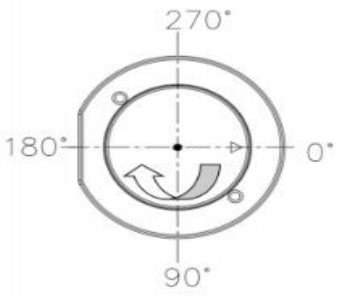
5.3 Product Dimensions



5.4 Scan Data Coordinate System Definition

The N10 series LiDAR uses a telescope-style optical structure with horizontally placed transmitter and receiver. When installing the LiDAR and designing robot system integration, the internal optical construction must be carefully considered for accurate effective detection angle design.

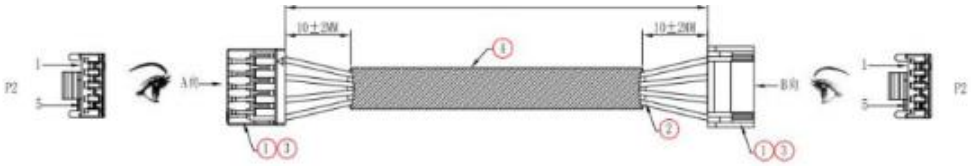
We define a polar coordinate system with the structural center of N10 as the pole, clockwise as positive direction, and the triangle marker position as the zero-degree angle.

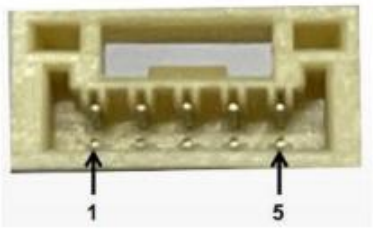


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6. Hardware Connection

The connector type on the LiDAR side: Male connector A1257H-5P; Female connector A1257WR-S-5P-LCP.





PIN 序	管脚	颜色	描述	最小值	典型值	最大值
1	GND	黑色	供电电压负极	0 V	0 V	0 V
2	TX	黄色	雷达数据输出	0 V	3.3 V	3.5 V
3	RX	绿色	功能控制脚	0 V	3.3 V	3.5 V
4	VCC	红色	供电电压正极	4.75 V	5 V	5.25 V
5	NetP1_1	蓝色	接机壳	-	-	-

The product includes a TTL to Type-C USB adapter module for TTL to USB data conversion, convenient for debugging. This adapter board is not a required component for LiDAR operation.



7. Instructions for Use

7.1 Preparations

Remove all protective films on the optical front window.

Prepare the matched TTL to Type-C USB adapter module and certified 5V power supply.

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Confirm the network environment: Windows PC / ROS environment ready.

7.2 Hardware Wiring

Connect the LiDAR body, SMH200-05H to MicroUSB adapter module, and MicroUSB cable as shown in the wiring diagram.

7.3 Windows System Operation

7.3.1 Driver Installation



WinPcap_4_1_3.zip

Before use, WinPcap dependency library must be installed first.

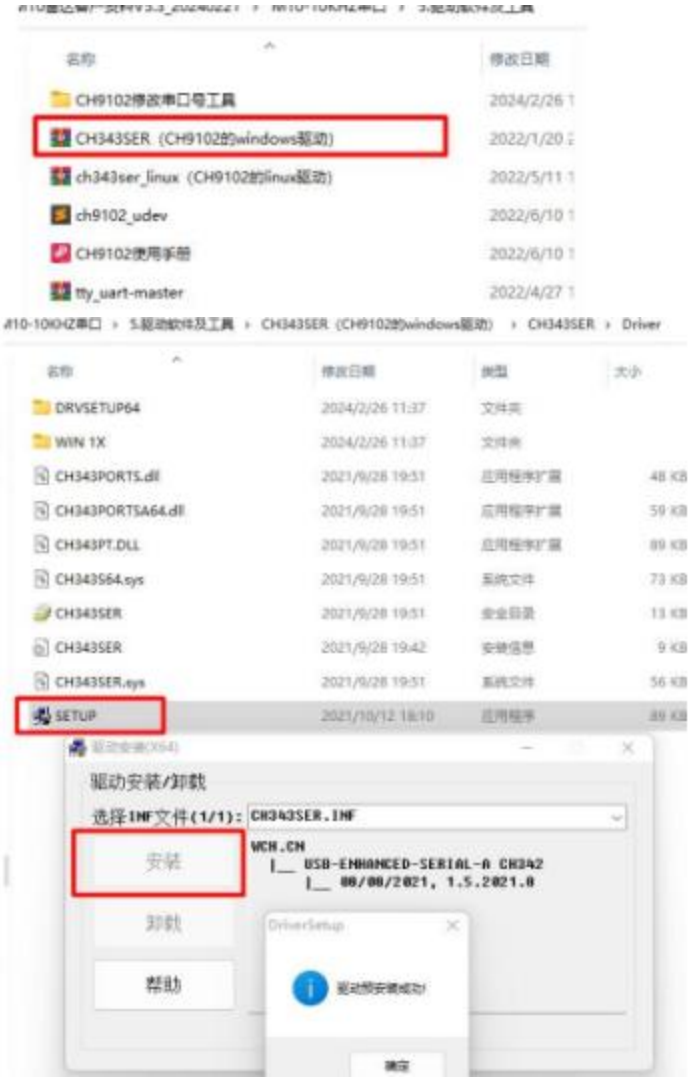


CH343SER（CH9102的windows驱动）.ZIP

Extract the CH343SER driver package from the data package and install it.

Double-click SETUP.EXE in the driver package to open, then click Install and wait for successful installation.

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7.3.2 Host Software Usage

Windows上位机.zip

1. Connect the USB cable to the computer USB port. In Windows, right-click This PC > Properties > Device Manager > Ports, find the CH9102 port name. The corresponding COM number is the LiDAR serial port.

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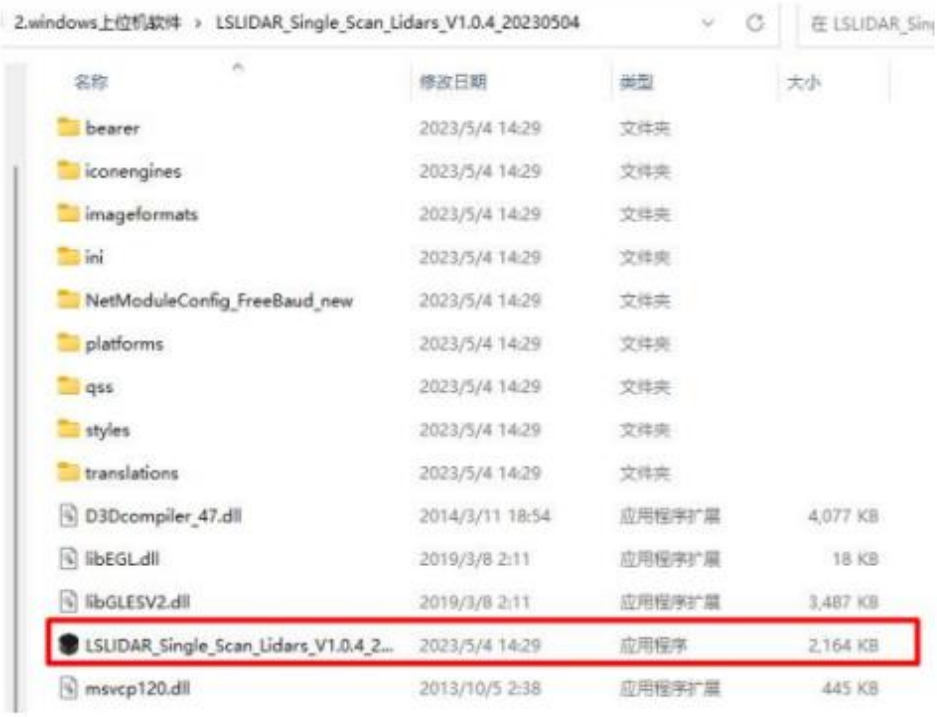


2. Open the extracted folder, find LSLIDAR_Single_Scan_Lidars.exe and double-click to run.



3. Click Auto Connect to connect to the LiDAR. If auto connection fails, select Manual Connect, choose the LiDAR model and corresponding serial port, then click OK.

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打开上位机软件后，可以点击自动连接便能连上雷达。



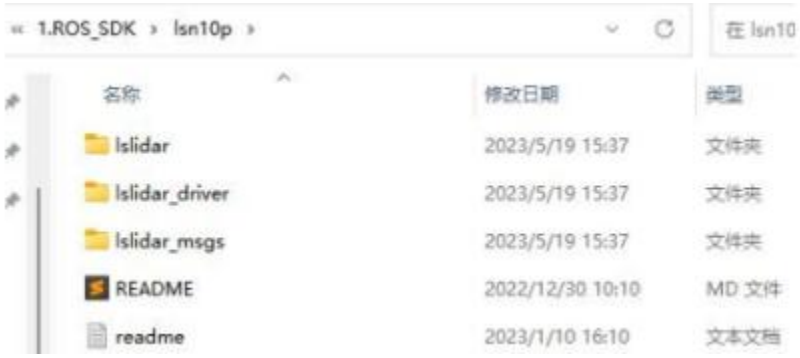
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7.4 ROS Environment Operation

7.4.1 Compile ROS SDK Package

1. Extract the ROS SDK compressed package to obtain the lsx10 package.

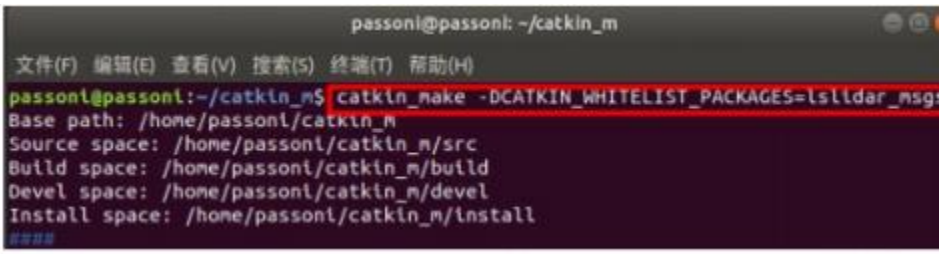


2. Copy the lsx10 package to the src directory of the ROS workspace.



3. Open a terminal and install dependencies: sudo apt-get install libpcap-dev

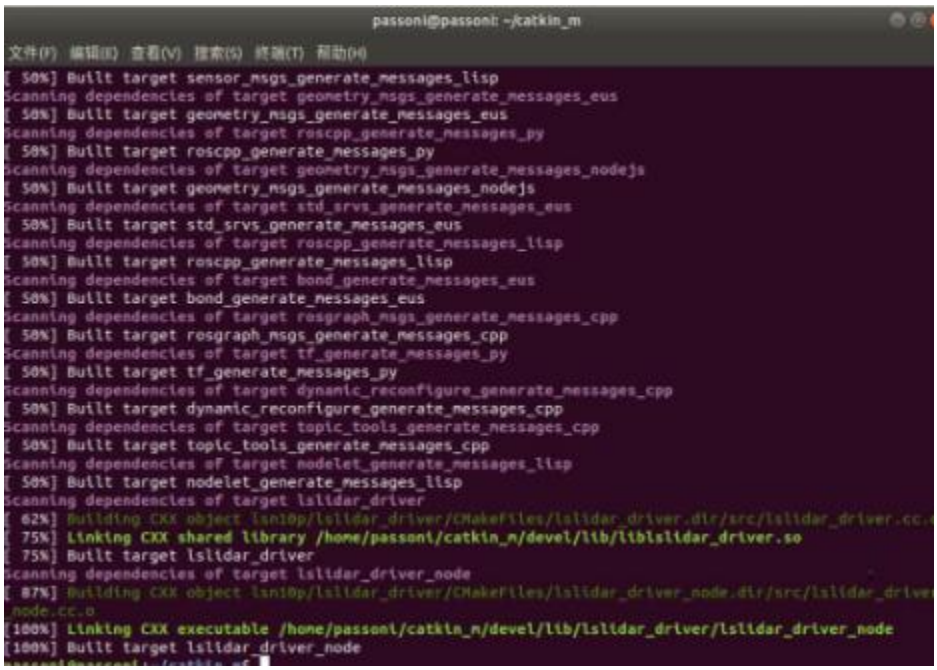
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4. Return to the workspace root directory and compile:

catkin_make -DCATKIN_WHITELIST_PACKAGES="lslidar_msgs"

catkin_make -DCATKIN_WHITELIST_PACKAGES="lslidar_driver"



7.4.2 Ubuntu CH9102 Driver Installation

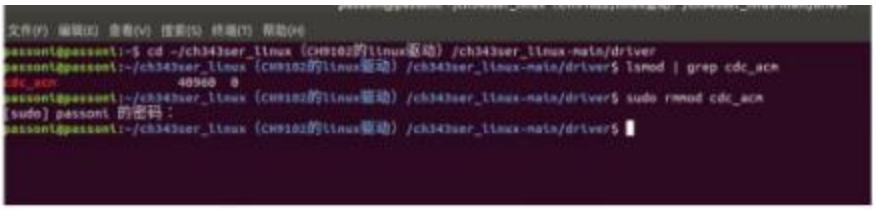
1. Check if the driver module exists: lsmod | grep ch343

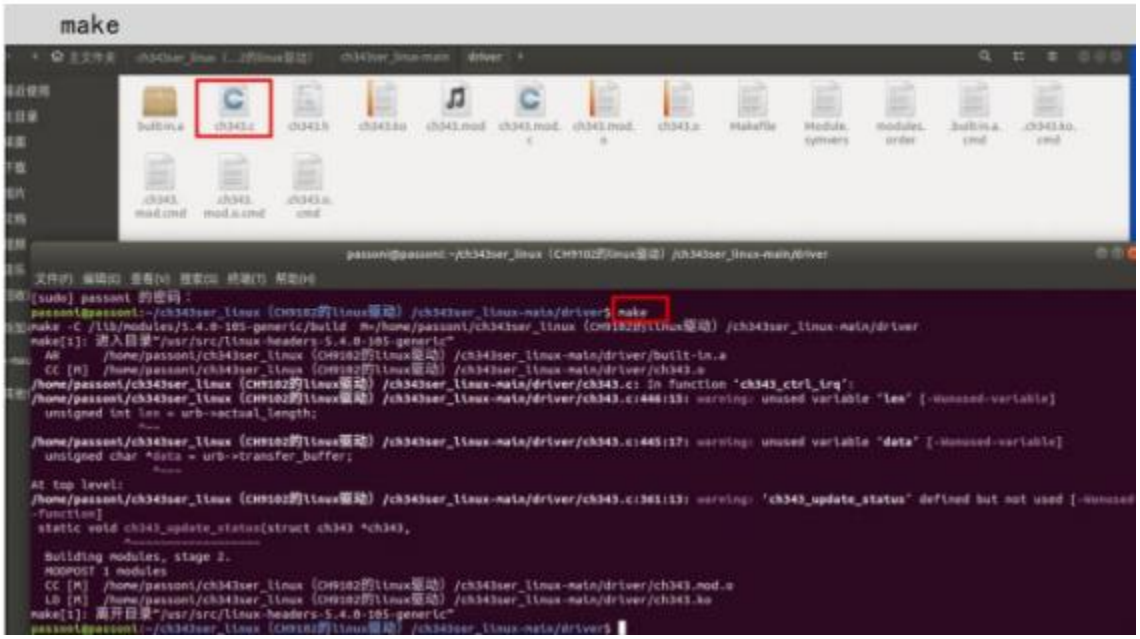


2. Extract the driver package.

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3. Remove cdc_acm module if loaded: sudo rmmod cdc_acm



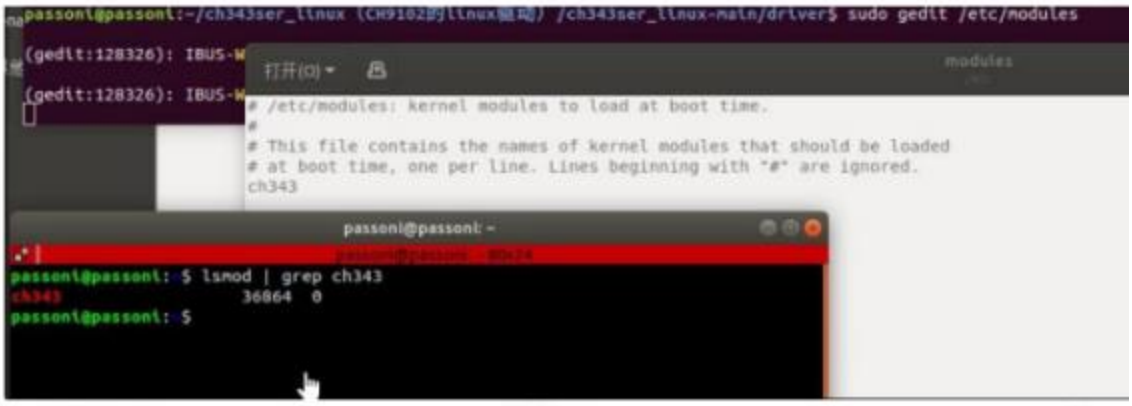


4. Compile: sudo make

5. Load module: sudo make load



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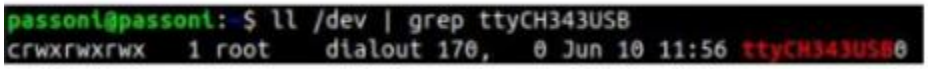


6. Install: sudo make install

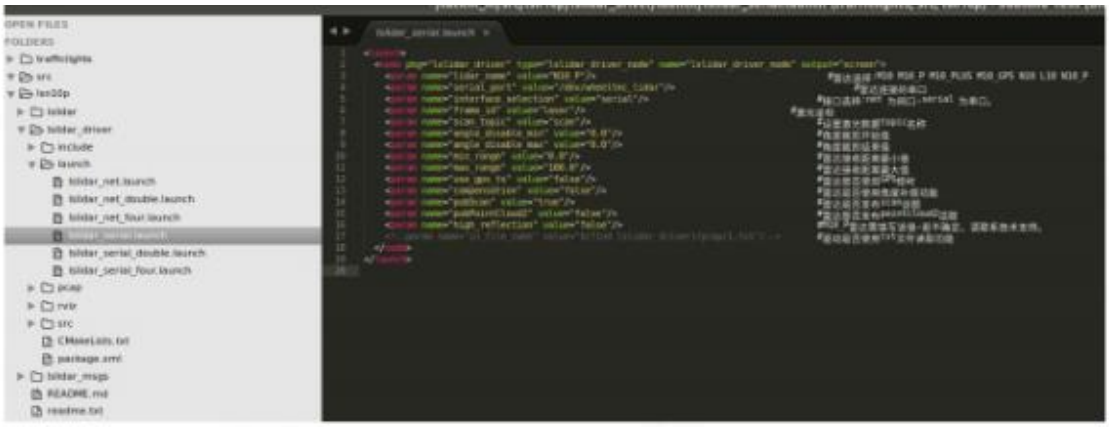
7. Add ch343 to /etc/modules for auto-load on boot: sudo gedit /etc/modules

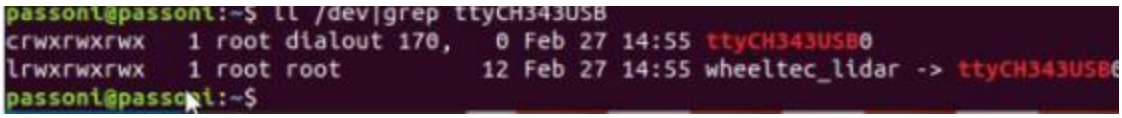
7.4.3 Using ROS Package to Drive N10 LiDAR

1. Connect the LiDAR to Ubuntu system. Check device recognition: ll /dev | grep ttyCH343USB



2. Open lslidar_serial.launch and configure the serial_port value to match the LiDAR serial port name.





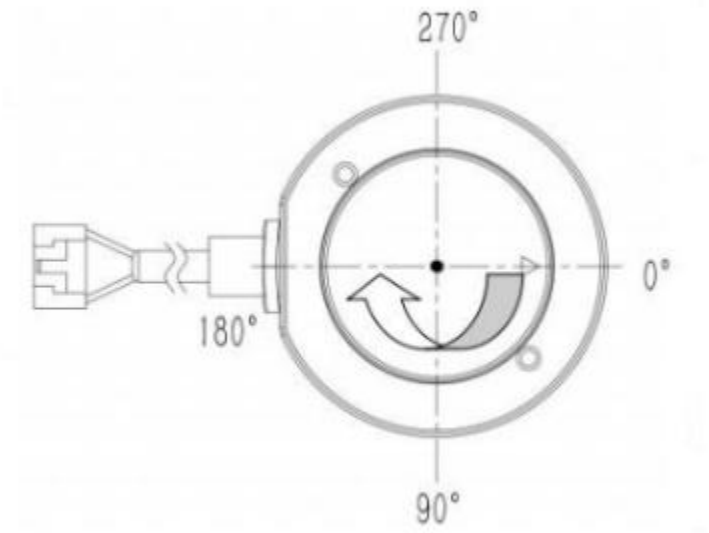
3. To shield specific angles: set truncated_mode=1, then configure disable_min and disable_max parameters.

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```
lslidar_serial.launch
<launch>
  <node pkg="lslidar_driver" type="lslidar_driver_node" name="lslidar_driver_node" output="screen">
    <param name="lidar_name" value="N10_P"/>                    #雷达选择:M10 M10_P M10_PLUS M10_GPS N10 L10 N10_P
    <param name="serial_port" value="/dev/wheeltec_lidar"/>          #雷达连接的串口
    <param name="interface_selection" value="serial"/>          #接口选择:net 为网口,serial 为串口。
    <param name="frame_id" value="laser"/>                   #激光坐标
    <param name="scan_topic" value="scan"/>                  #设置激光数据topic名称
    <param name="angle_disable_min" value="0.0"/>            #角度裁剪开始值
    <param name="angle_disable_max" value="0.0"/>            #角度裁剪结束值
    <param name="min_range" value="0.0"/>                    #雷达接收距离最小值
    <param name="max_range" value="100.0"/>                  #雷达接收距离最大值
    <param name="use_gps_ts" value="false"/>                 #雷达是否使用GPS授时
    <param name="compensation" value="false"/>               #雷达是否使用角度补偿功能
    <param name="pubScan" value="true"/>                     #雷达是否发布scan话题
    <param name="pubPointCloud2" value="false"/>             #雷达是否发布pointcloud2话题
    <param name="high_reflection" value="false"/>            #M10_P雷达需填写该值,若不确定,请联系技术支持。
    <!--param name="in_file_name" value="$(find lslidar_driver)/pcap/1.txt"/-->  #驱动是否使用txt文件读取功能
  </node>

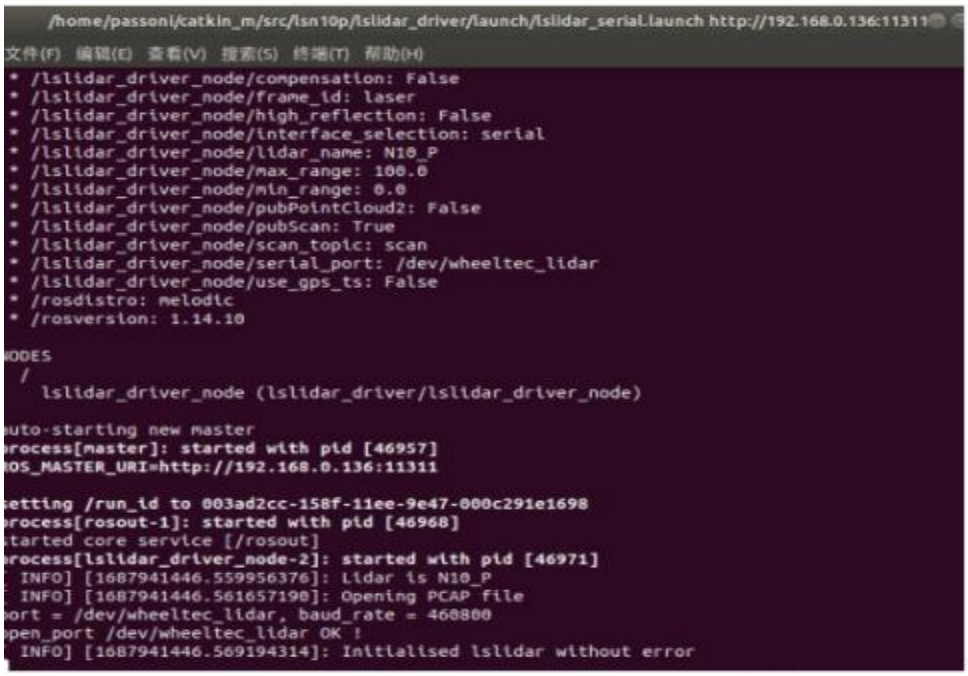
      <!-- 镭神N10P雷达参数 -->
    <param name="lslidar_driver_node/truncated_mode" value="0"/>    <!--0:不限角度 1:限角度-->
    <rosparam param="lslidar_driver_node/disable_min">[120]</rosparam> <!--角度左值-->
    <rosparam param="lslidar_driver_node/disable_max">[240]</rosparam> <!--角度右值-->
    <!-- 镭神N10P雷达参数 -->

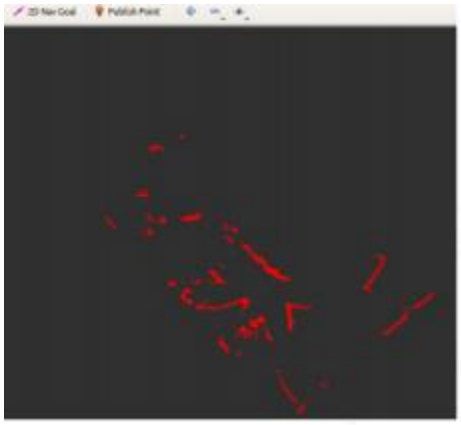
</launch>
```

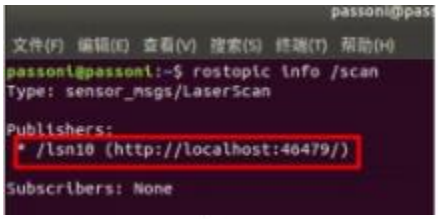


4. Launch the LiDAR: roslaunch lslidar_driver lslidar_serial.launch

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8. COMPLIANCE INFO

This electronic LiDAR product must print CE mark at the end of manual, CE mark height >= 5mm.

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Compliance Certifications: CE, RoHS, UKCA

Applicable EU Directives: LVD 2014/35/EU, EMC 2014/30/EU, WEEE 2012/19/EU, RoHS 2011/65/EU

Laser Standard: IEC 60825-1



CE: Conformite Europeenne, this marking confirms the product complies with applicable EU directives (LVD 2014/35/EU, EMC 2014/30/EU, IEC 60825-1 Laser Safety Standard).



UKCA: United Kingdom Conformity Assessed, compliant with UK market regulations.

9. MAINTENANCE & STORAGE

Cleaning: Power off LiDAR completely before cleaning. Wipe optical window with soft lint-free dry cloth only.

Daily Storage: Store in dry indoor environment within operating temperature range.

Long-term Storage: If unused over 1 month, disconnect power and store within -30 to 70 degrees C.

10. DISPOSAL & RECYCLING

This symbol indicates this electronic LiDAR cannot be discarded with household waste. According to WEEE Directive (2012/19/EU), send the product to designated electronic waste recycling stations.

11. FAQ

Q1: What is the baud rate of the N10 LiDAR?

The baud rate is 230400bps.

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12. Contact Us



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